2026 MIDDLE SCHOOL STEM CAMPS

From Trees to Tech

INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

IN THIS PACKET

MudWatt Bioelectric League

Turn a cup of mud into a living battery

At a glance: Can your team make mud generate electricity and defend your design with data? About 80 min; 4 teams.

MATERIALS

- 4 DIY soil microbial fuel cells: clear deli container, pond mud, two 4×4 in carbon-felt electrodes each (plus 1 spare set)
- measured sugar cubes as fuel (one per cell per run, plus spares)
- 4 multimeters, one per team, plus 2 spare (confirm battery type and that a battery is included)
- 100 kΩ resistors and alligator-clip lead wires, one set per cell
- nitrile gloves and a containment tray per station
- labels and masking tape
- handwashing supplies

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which design change moved your readings the most, and how do you know?
- Why must the anode stay buried and the cathode stay in air?
- How is your weekly log like the work of an energy engineer?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage reading	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	**100**

SOURCES soil microbial fuel cell electrode and voltage references (carbon-felt anodes; terrestrial MFC literature on 300 to 600 mV output as the biofilm matures).

Forest Sensor Sprint

Site the smartest place for a campus tree sensor

At a glance: Find the smartest place to install a future campus tree sensor. About 80 min; 4 teams.

MATERIALS

- 4 clipboards plus spares
- 4 paper clinometers plus spares
- 4 combined temperature + humidity meters (one device, plus spares)
- 4 three-in-one soil meters (moisture, light, pH), one per team, plus spares
  - 1 to 2 long tape measures (pre-mark a fixed standoff distance in prep)
- 4 route cards (one per team) plus 2 spares
- pencils

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

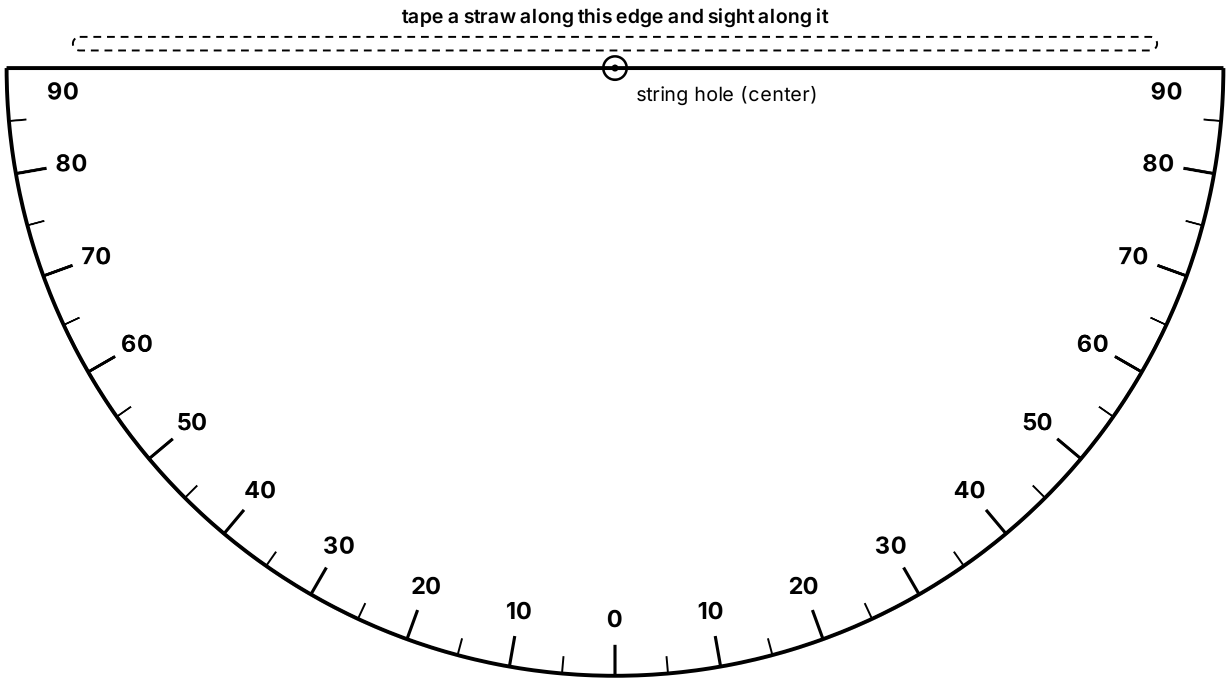
- Which measurement most influenced your choice?
- Where did two nearby spots disagree, and why?
- What would change your recommendation in winter?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Data accuracy	30
Route efficiency	20
Site recommendation evidence	30
Teamwork and field conduct	20
Total	**100**

SOURCES GLOBE Observer Trees; NASA MyNASAData Urban Heat Island

Paper clinometer



Cut along the flat top edge and the curved outline.

ASSEMBLE

01 Cut out the half circle along the curved outline.

02 Tape a straight straw along the top straight edge.

03 Punch the center dot, thread a string through it, and tie a washer or paper clip on the end as a weight.

04 Sight the treetop through the straw, let the string hang free, then pinch it against the scale and read the number.

05 Self-check: sight something level (reads 0) and straight up (reads 90).

FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor marked.

ANGLE	TAN
10	0.18
20	0.36
30	0.58
40	0.84
45	1.00
50	1.19
60	1.73
70	2.75

Example: eye height 1.5 m, distance 10 m, angle 40° gives 1.5 + 10 × 0.84 = about 9.9 m.

Team route card

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team ______________________   Date _____________

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (RELATIVE)	SOIL MOISTURE (1 TO 10)	NOTES
Reference (calibrate)						
A						
B						
C						
D						
E						

Our recommended sensor site: ____

Why (use your data): ____

The checkpoints are the spots the instructor marked outside.

cut here

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team ______________________   Date _____________

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (RELATIVE)	SOIL MOISTURE (1 TO 10)	NOTES
Reference (calibrate)						
A						
B						
C						
D						
E						

Our recommended sensor site: ____

Why (use your data): ____

The checkpoints are the spots the instructor marked outside.

INSTRUCTOR · TTT-03 · BIOMIMICRY

Seed Dispersal Derby

Engineer a seed that escapes its parent tree

At a glance: Build a seed that escapes the parent tree without engines or throwing. About 80 min; 4 teams.

MATERIALS

- paper
- cardstock
- paper clips
- masking tape
- string
- fan or marked test lane
- timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which change improved hang time without losing distance?
- Why does spinning slow the fall?
- Which tree's strategy did your final seed copy?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	**100**

SOURCES Science Buddies Seed Dispersal; Project Learning Tree

Xylem Pipeline Relay

Move water uphill the way a tree does

At a glance: Move water uphill like a tree, faster and cleaner than rival teams. About 80 min; 4 teams.

MATERIALS

- paper towels
- cotton string
- felt strips
- cups
- food coloring
- trays
- rulers
- elevation blocks

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which material moved water best, and why?
- How did slope change your delivery?
- Where does a real tree use this same effect?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	**100**

SOURCES Science Buddies; plant water transport references

Greenhouse Climate Controller

Match plants to the right climate zone

At a glance: Run the greenhouse controls for a plant that cannot complain, only wilt. About 80 min; 4 teams.

MATERIALS

- plant profile cards
- control dial boards
- clipboards
- dry erase markers
- tour clue sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	**100**

SOURCES Temple Ambler Greenhouse

Pinecone Weather Machine

A humidity sensor with no battery and no code

At a glance: Build a weather sensor with no battery and no code. About 80 min; 4 teams.

MATERIALS

- dry pine cones
- index cards
- wood skewers
- spray bottles
- rulers
- masking tape
- paper fasteners

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does a two-layer strip bend when only one layer swells?
- What made your indicator faster?
- Where would a no-power humidity sensor be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Response accuracy	35
Response speed	20
Readable output scale	20
Biomimicry explanation	15
Redesign quality	10
Total	**100**

SOURCES AskNature Pine Cones Strategy

Pollinator Network Draft and Build

Design a habitat that works all season

At a glance: Draft a pollinator habitat that works all season, not just on the prettiest day. About 80 min; 4 teams.

MATERIALS

- plant cards
- pollinator cards
- grid boards
- stickers
- markers
- season tokens

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Where was the biggest gap in your bloom calendar?
- Why does clumping help pollinators?
- What happens to the network if one season is missing?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Seasonal coverage	30
Pollinator fit	25
Native and clumping logic	20
Layout clarity	15
Defense	10
Total	**100**

SOURCES USDA Pollinators; US Forest Service pollinator guidance

Arboretum Eco-Quest

Solve outdoor ecology checkpoints

At a glance: Read the living collection and solve the checkpoints before time runs out. About 80 min; 4 teams.

MATERIALS

- clue cards
- route map
- clipboards
- pencils
- evidence token bags
- optional QR codes

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	**100**

SOURCES Temple Ambler Arboretum; Arboretum Explorer

Minecraft Tree World Resilience Cup

Design a climate-resilient landscape

At a glance: Design a landscape that survives stress because of how it is built, not luck. About 80 min; 4 teams.

MATERIALS

- Minecraft Education devices and template or paper grid kit
- rubric cards
- display timer

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which feature would help most in a real storm?
- Why does diversity add resilience?
- What in your build would fail first, and how would you fix it?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Resilience features	30
Biodiversity support	25
Realism	20
Explanation	15
Build polish	10
Total	**100**

SOURCES Minecraft Education Biodiversity

Tree Ring Climate Detective

Read rings to reconstruct past climate

At a glance: Read the rings to reconstruct the climate events a tree lived through. About 80 min; 4 teams.

MATERIALS

- printed ring cards
- wood cookies or images
- answer boards
- timers
- claim-evidence cards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which pattern was easiest to misread?
- How is a tree ring like and unlike a thermometer?
- What other natural records store past climate?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	**100**

SOURCES NOAA Tree Rings; EPA Tree Rings, Living Records of Climate

Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

At a glance: Build a surface that sheds water and dirt the way a lotus leaf does. About 80 min; 4 teams.

MATERIALS

- wax paper
- cardstock
- sandpaper strips
- masking tape
- pipettes
- water
- pepper or cocoa residue
- ramps

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	**100**

SOURCES AskNature Lotus Leaf Self-Cleaning Surface

Leaf Stomata Microscope Detective

Count stomata and rank leaves by water strategy

At a glance: Count the breathing pores and rank leaves by how they manage water. About 80 min; 4 teams.

MATERIALS

- leaf samples
- clear tape
- clear nail polish or prepared slides
- USB microscopes or magnifiers
- grid sheets
- gloves

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why count several fields instead of one?
- What does a high stomata count suggest about a leaf's habitat?
- Where might error sneak into your count?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Counting accuracy	30
Ranking evidence	25
Team data table	10
Cleanup and care	10
Total	**100**

SOURCES Leaf stomata microscopy adaptations; Science Buddies plant biology

BACKUP ACTIVITIES

Ready alternates

Root Grip Erosion Rescue

Can roots hold a slope together in a storm?

At a glance: Can roots hold a slope together during a storm? About 60 min; 4 teams.

MATERIALS

- Trays
- Soil
- String or yarn roots
- Craft sticks
- Watering cups

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why did the rooted slope hold more soil?
- How would deeper roots change the result?
- Where is this used to protect real hillsides?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Soil retained	40
Runoff control	25
Design explanation	20
Cleanup	15
Total	**100**

SOURCES EPA green infrastructure and erosion lessons

INSTRUCTOR · TTB-02 · FIELD SCIENCE

Tree Height Triangulation Shootout

Measure a tree without climbing it

At a glance: Measure a tree without climbing it. About 60 min; 4 teams.

MATERIALS

- Paper clinometers
- String and washers
- Rulers and tape measures
- Clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- What was your biggest source of error?
- Why does eye height matter in the calculation?
- Where is remote height measurement used in the field?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	**100**

SOURCES GLOBE Trees Activities

Urban Heat Shade Dash

Find the coolest route through a hot campus

At a glance: Find the coolest route through a hot campus. About 60 min; 4 teams.

MATERIALS

- Thermometers or IR thermometers
- Campus maps
- Clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- How big was the sun-to-shade temperature gap?
- What surface was hottest, and why?
- How do cities use shade to cool streets?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Cool route design	35
Data quality	25
Reasoning	20
Communication	20
Total	**100**

SOURCES NASA MyNASAData Urban Heat Island

Photosynthesis Float-Off Playoffs

Make leaf disks float by making oxygen

At a glance: Make leaf disks float by producing oxygen. About 60 min; 4 teams.

MATERIALS

- Spinach leaves
- Baking soda solution
- Cups
- Oral syringes, no needle
- Timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does oxygen make the disks float?
- Which variable changed the rate most?
- How is this evidence of photosynthesis?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Controlled-variable design	25
Data table	20
Explanation	10
Cleanup	10
Total	**100**

SOURCES Exploratorium Photosynthetic Floatation